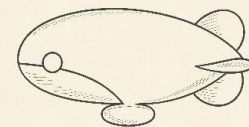


The Evolution of the Hung Task Detector

Making Hung Task reports blocker-aware

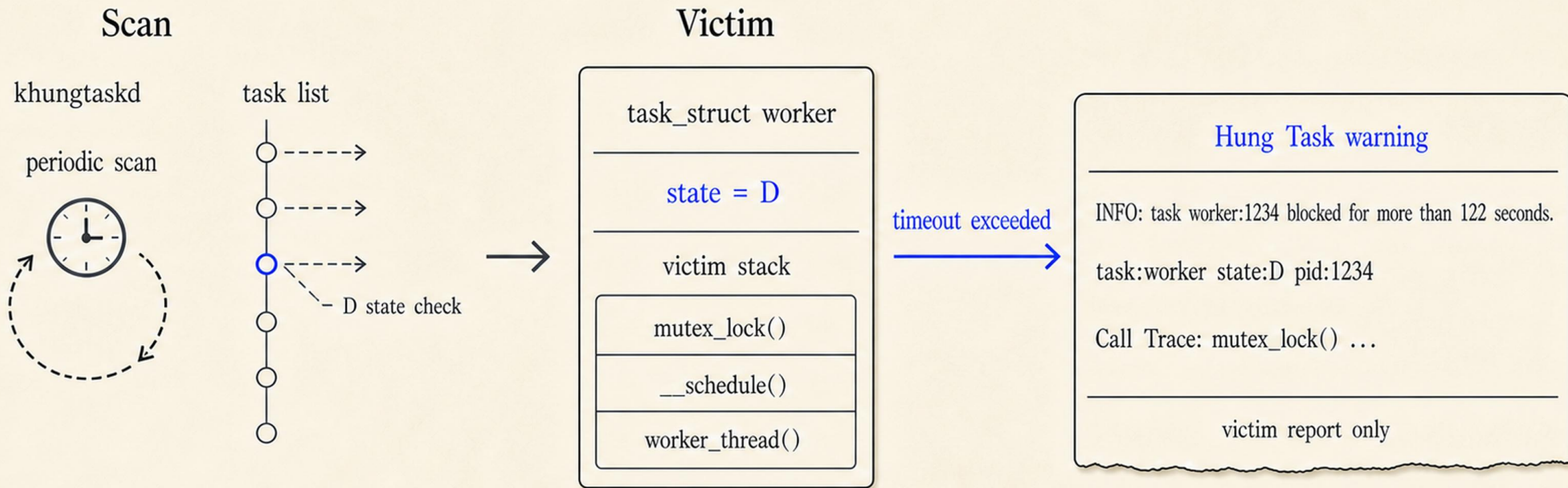
Masami Hiramatsu Lance Yang Zi Li

Hung Task starts with the victim



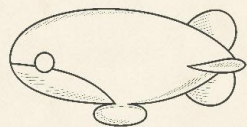
khungtaskd turns a long D state wait into a report

Linux kernel / mutex / semaphore / rwsem



The first report names the stuck task; the next slide shows the missing hop.

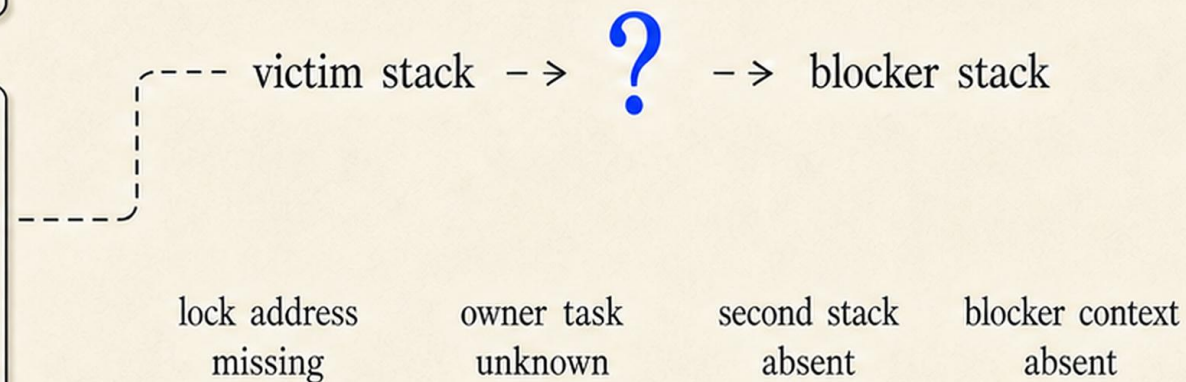
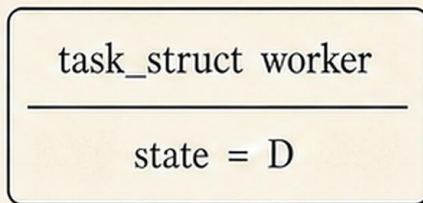
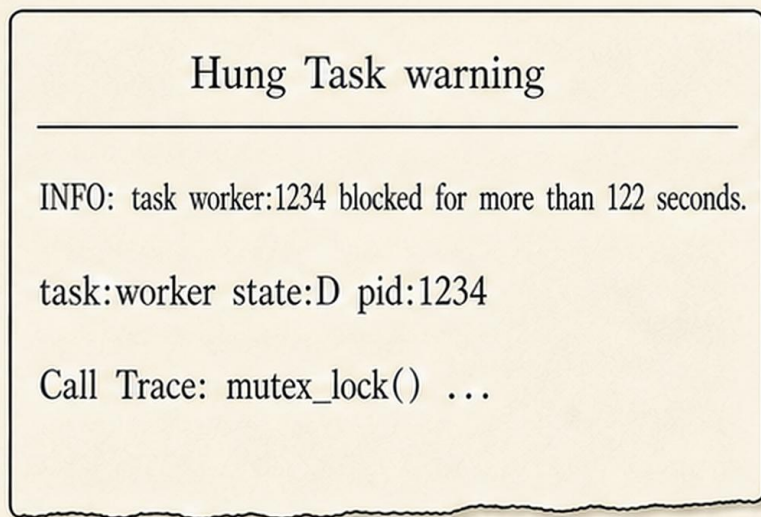
The old report stops at the victim



Why a victim stack alone does not answer the real question

Linux kernel / mutex / semaphore / rwsem

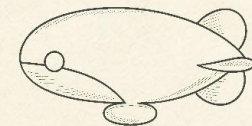
visible old report



Manual triage: infer lock from wait site, then verify by subsystem evidence

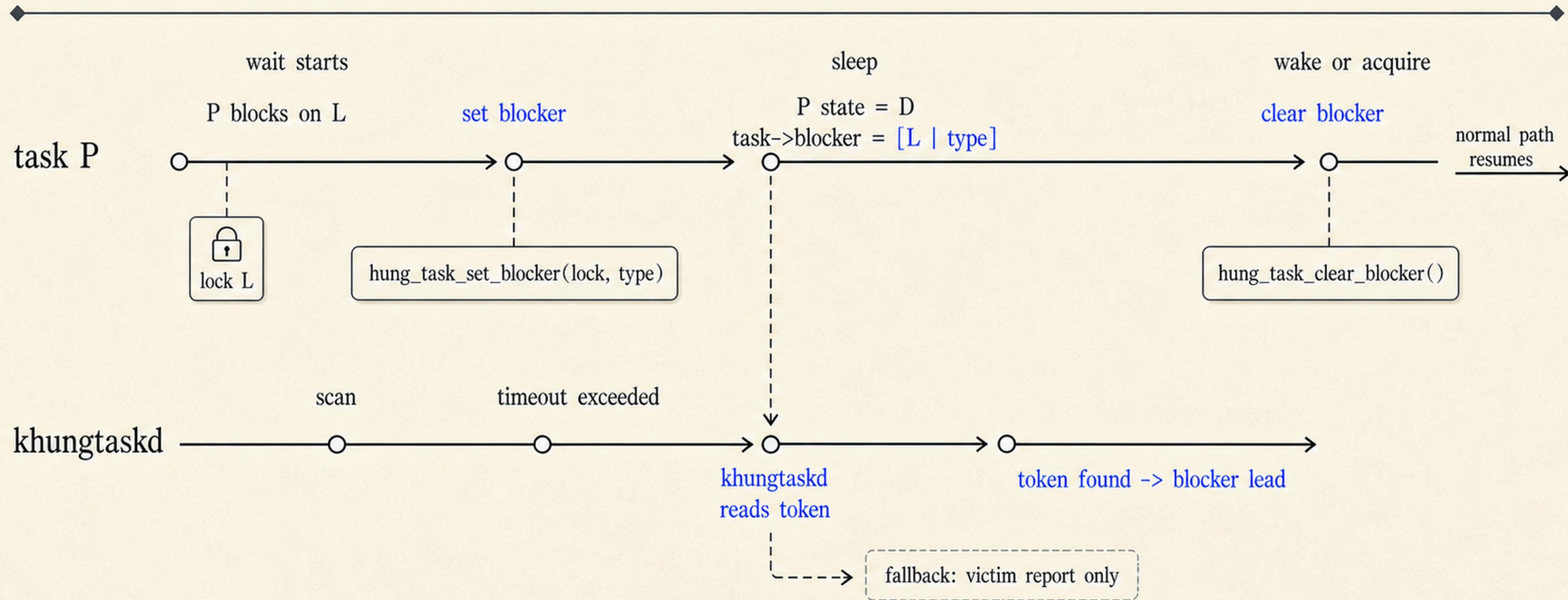
The old report stops at the first question; debugging wants the next hop.

Leave a token at the sleep point



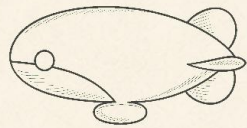
When the blocked task records the lock it waits on

Linux kernel / mutex / semaphore / rwsem



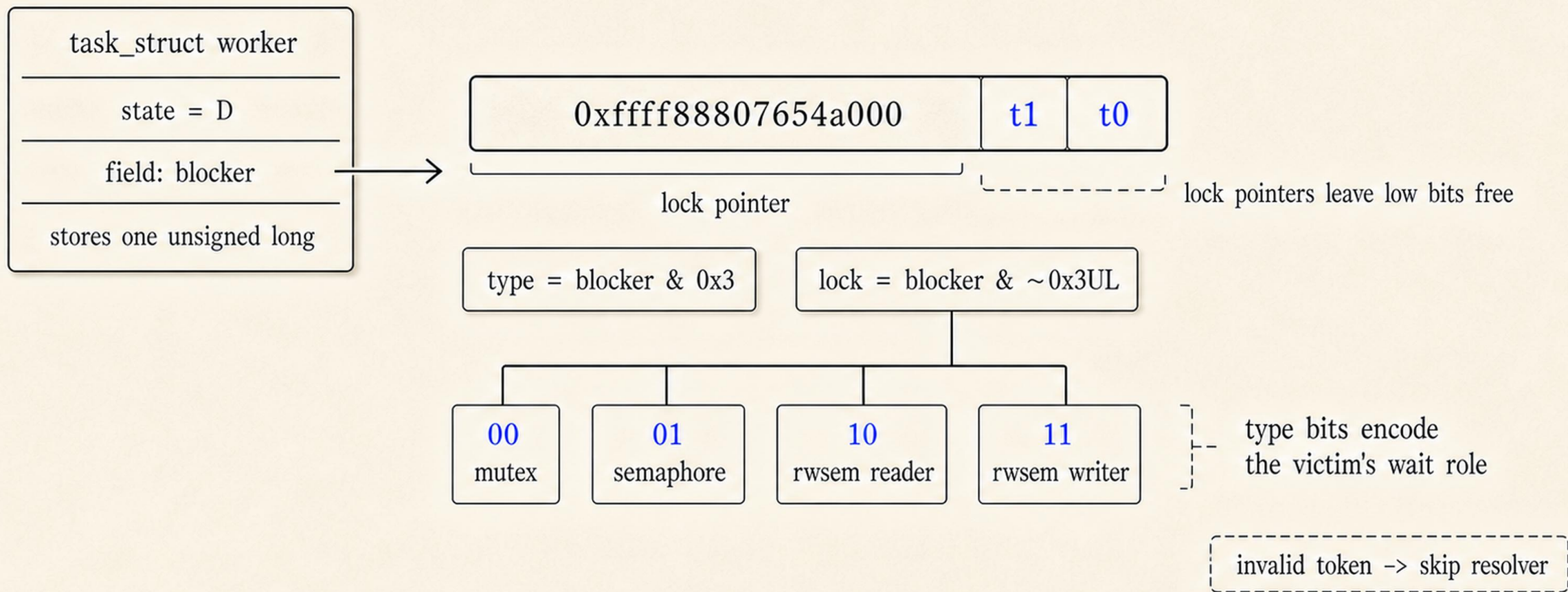
Left before sleep, cleared on wake; the token only covers the suspicious window.

One word: lock pointer plus type bits



One unsigned long holds a lock pointer and a resolver type

Linux kernel / mutex / semaphore / rwsem



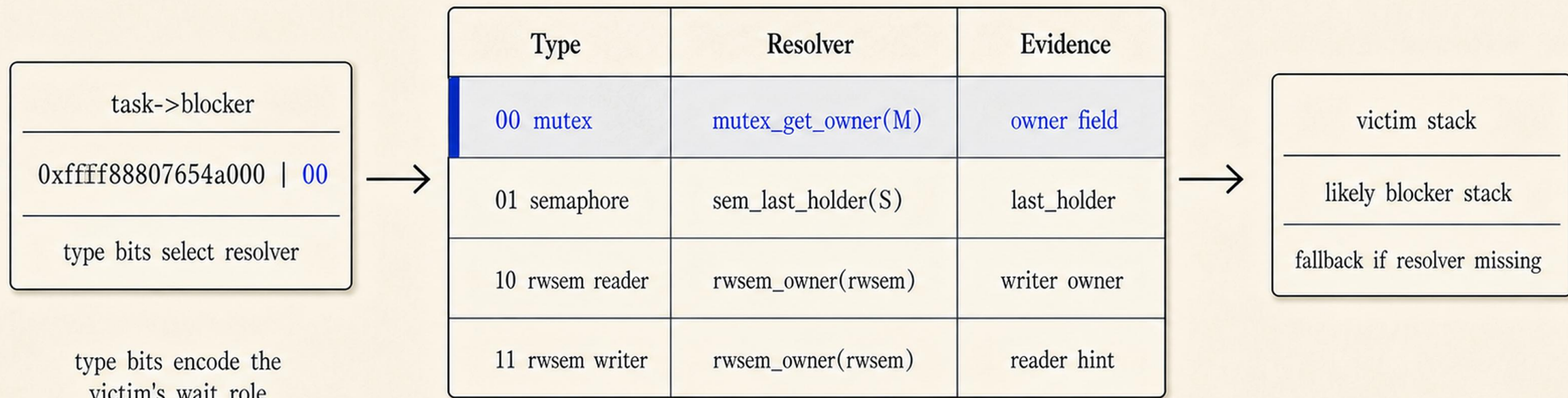
The pointer finds the lock object; the low bits choose the resolver path.

The token selects the evidence path



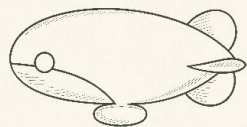
Decoded type bits dispatch a lock-specific evidence reader

Linux kernel / mutex / semaphore / rwsem



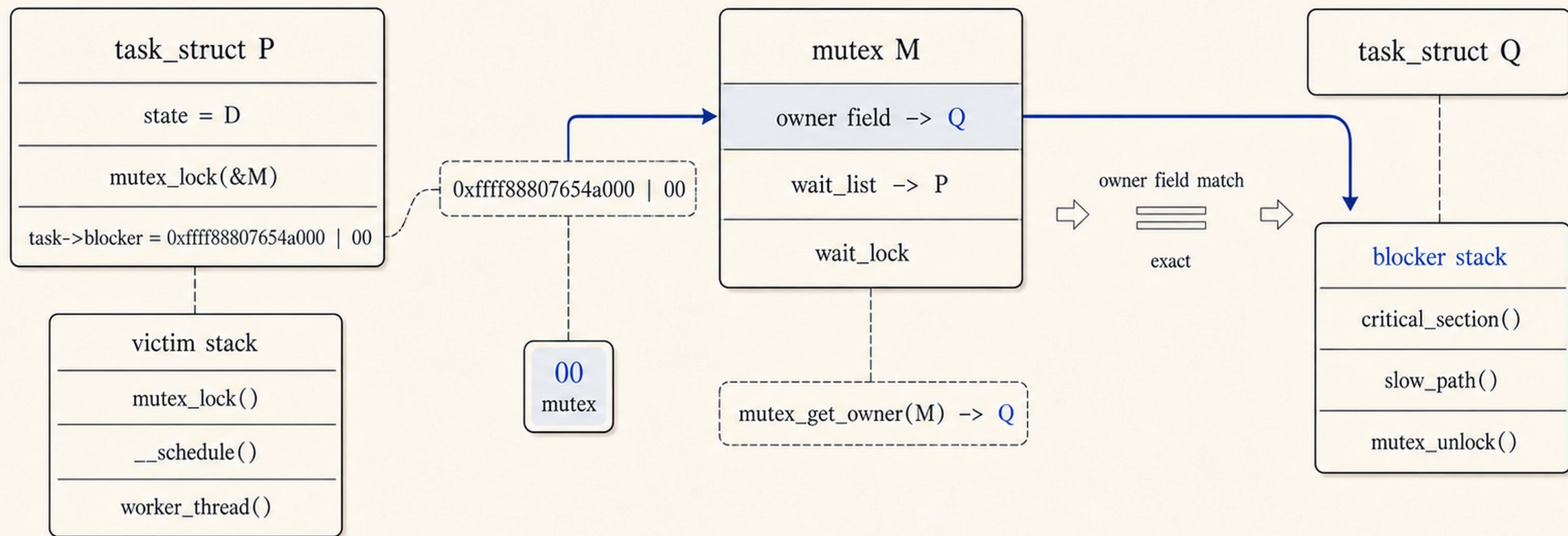
Same token, different locks; each resolver reads a different evidence field.

Mutex: the clean owner hop



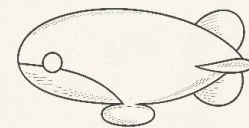
Mutex has one owner field, so the route reaches the owner

Linux kernel / mutex / semaphore / rwsem



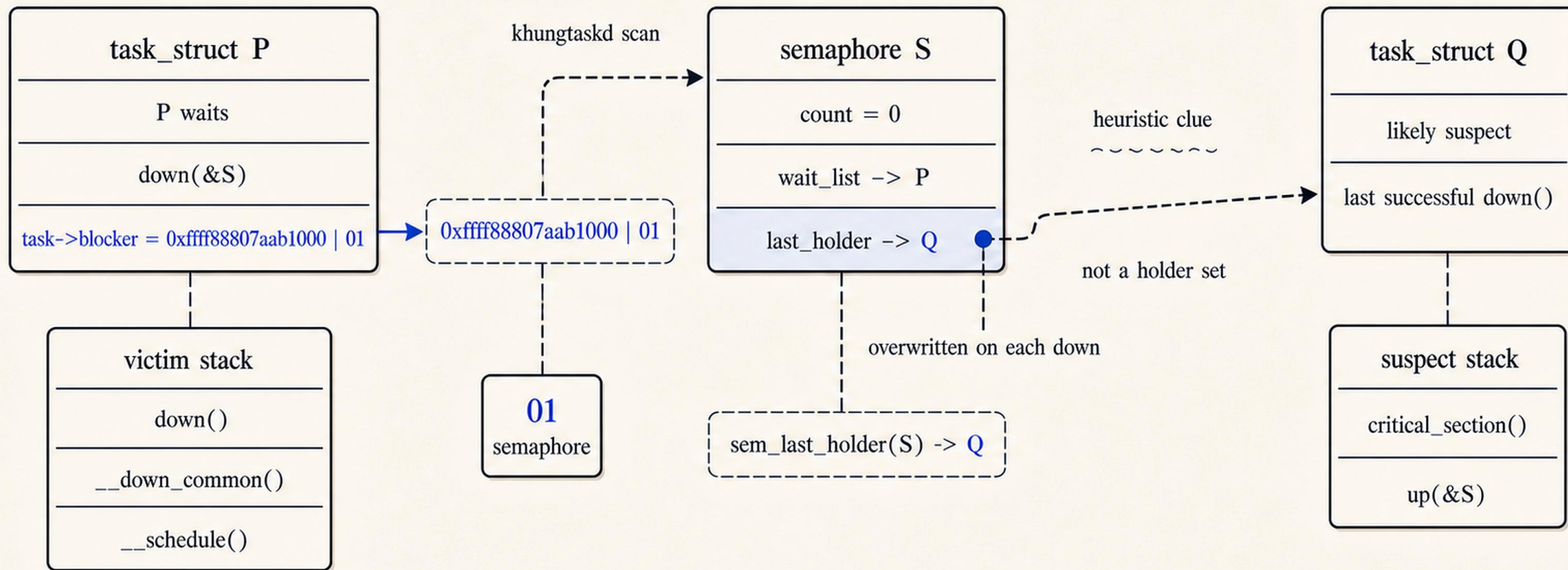
The clean baseline: the victim waits on the lock, and the owner is the next hop.

Semaphore: a breadcrumb, not an owner



No owner field, so the last acquirer is a lead, not an owner

Linux kernel / mutex / semaphore / rwsem



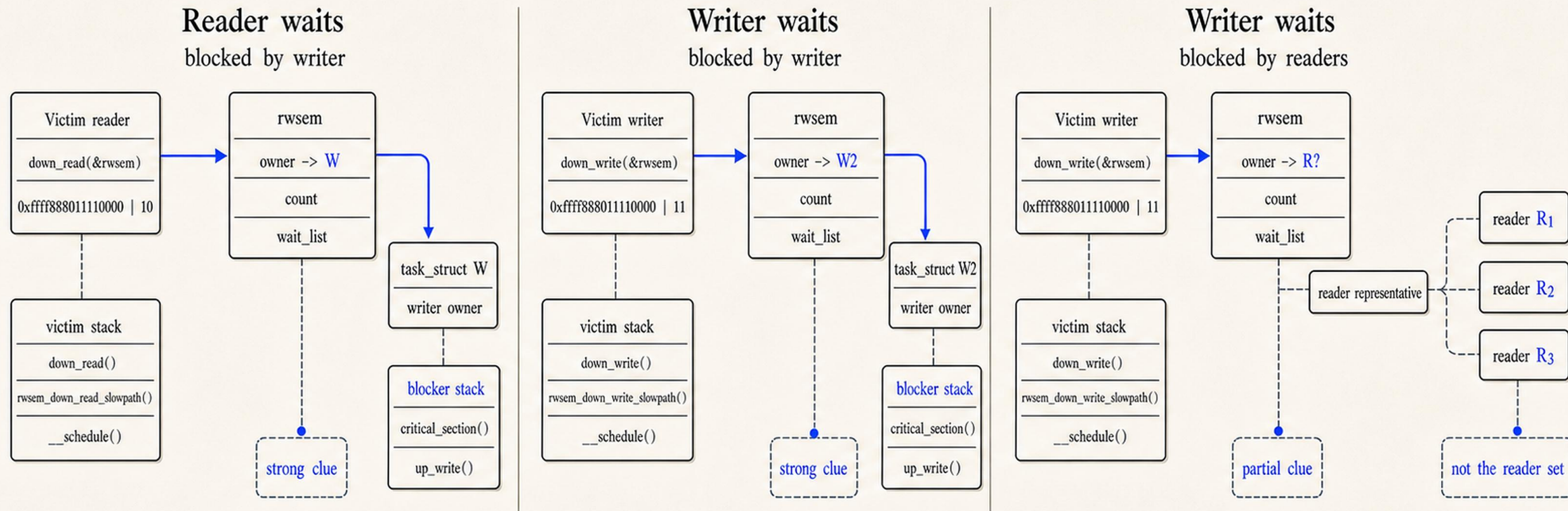
No real owner here; last_holder is a suspect, not a verdict.

Rwsem: asymmetric evidence



Reader-wait and writer-wait expose different evidence

Linux kernel / mutex / semaphore / rwsem



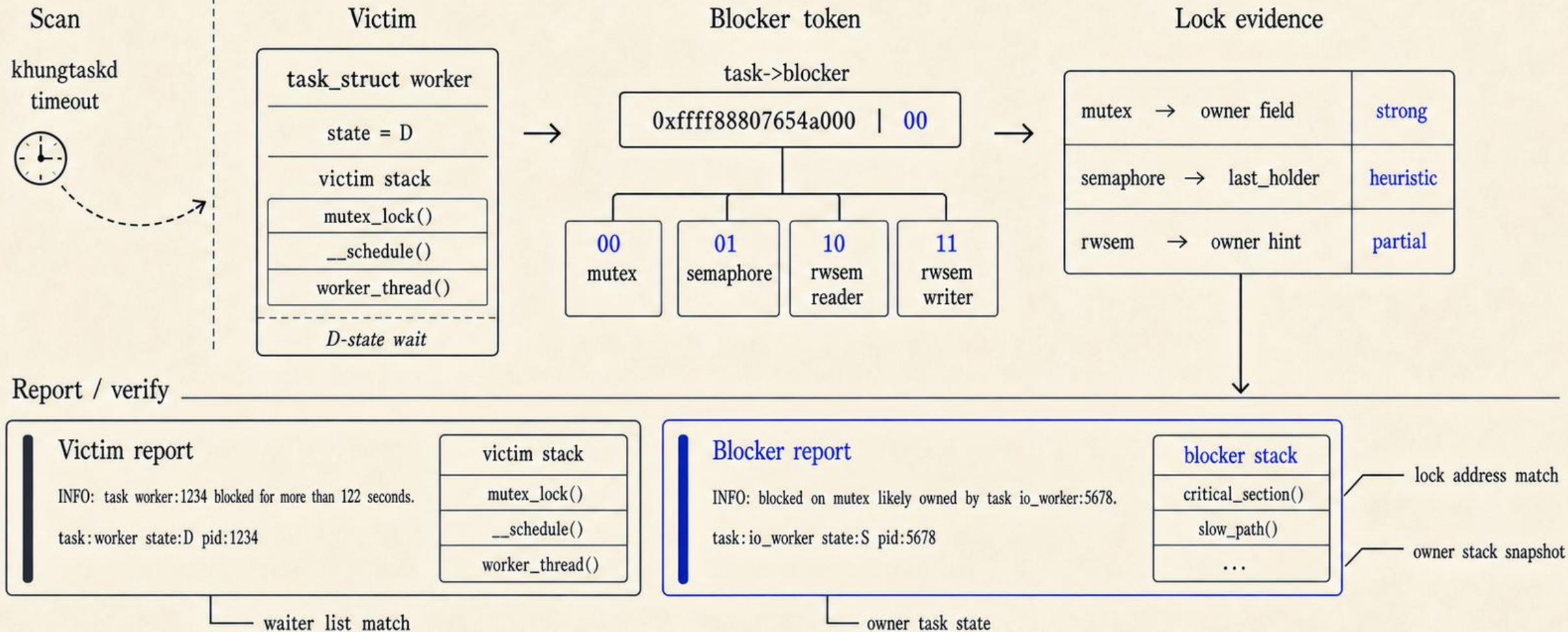
Strong whenever owner names a writer; partial only when owner names a reader representative.

From hung victim to blocker stack



A diagnostic map for Linux Hung Task blocker tracking

Linux kernel / mutex / semaphore / rwsem



Today's map answers not who is stuck, but who is likely blocking it.